

HYDROFARMS AGRI SOLUTIONS

SERVICE GREENHOUSE  
GERMINATION, GROW LIGHTING AND PACKING

Installation, Hygiene, Testing and Commissioning Method Statement

| Document field     | Controlled value                                                        |
|--------------------|-------------------------------------------------------------------------|
| Document code      | HF-GH01-MST-SRV-001                                                     |
| Revision           | 00 - Draft for coordination                                             |
| Project            | One-Feddan NFT Lettuce Greenhouse Project                               |
| Location           | Alexandria, Egypt                                                       |
| Service greenhouse | 9 x 36 m; germination north / packing south                             |
| Germination basis  | 90 operating trays; 100 procured 200-cell trays; 18,000 seedlings/batch |
| Prepared by        | Hydrofarms Agri Solutions                                               |
| Issue date         | 15 August 2026                                                          |

| LATEST APPROVED BASIS                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The germination tank is 1 m3. The expression 200 eyes means 200-cell propagation trays; it does not mean 200 trays. Earlier 500-litre and 200-tray study values are superseded. |

## 1. Purpose, Scope and Functional Boundaries

This method defines coordinated installation and commissioning of the northern germination area, rack grow lighting and southern dry-packing area inside the service greenhouse. It covers equipment placement, hygienic workflow, local water/mist distribution, lighting performance, drainage interfaces, testing, training and records.

| Area/interface                                                                                                                                                                                          | Approved basis         | Execution requirement                                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Service greenhouse                                                                                                                                                                                      | 9 x 36 m               | West of production greenhouse; full concrete floor; north side faces cooling pads and south side faces exhaust fans.           |
| Germination zone                                                                                                                                                                                        | Northern half          | Rack-based seedling production located toward the cooling pads.                                                                |
| Packing zone                                                                                                                                                                                            | 9 x 17 m southern area | Dry receiving, grading, weighing and packing area located beside the exhaust-fan wall.                                         |
| Protected movement                                                                                                                                                                                      | 2.5 m central corridor | Anti-virus-screen protected route connecting the external 4 x 4 m disinfection room and the 2.5 m sliding door to production.  |
| Environmental system                                                                                                                                                                                    | Pad-and-fan cooled     | Three service exhaust fans, separate cooling pads and two 600 mm circulation fans; no sealed air-conditioned germination room. |
| <b>BOQ OWNERSHIP</b>                                                                                                                                                                                    |                        |                                                                                                                                |
| Racks, trays, local germination water package and packing equipment are described here. Grow lights, feeders, controls, civil, covering and cooling retain their own BOQ ownership and are priced once. |                        |                                                                                                                                |

## 2. Reference Layout and Material Flow

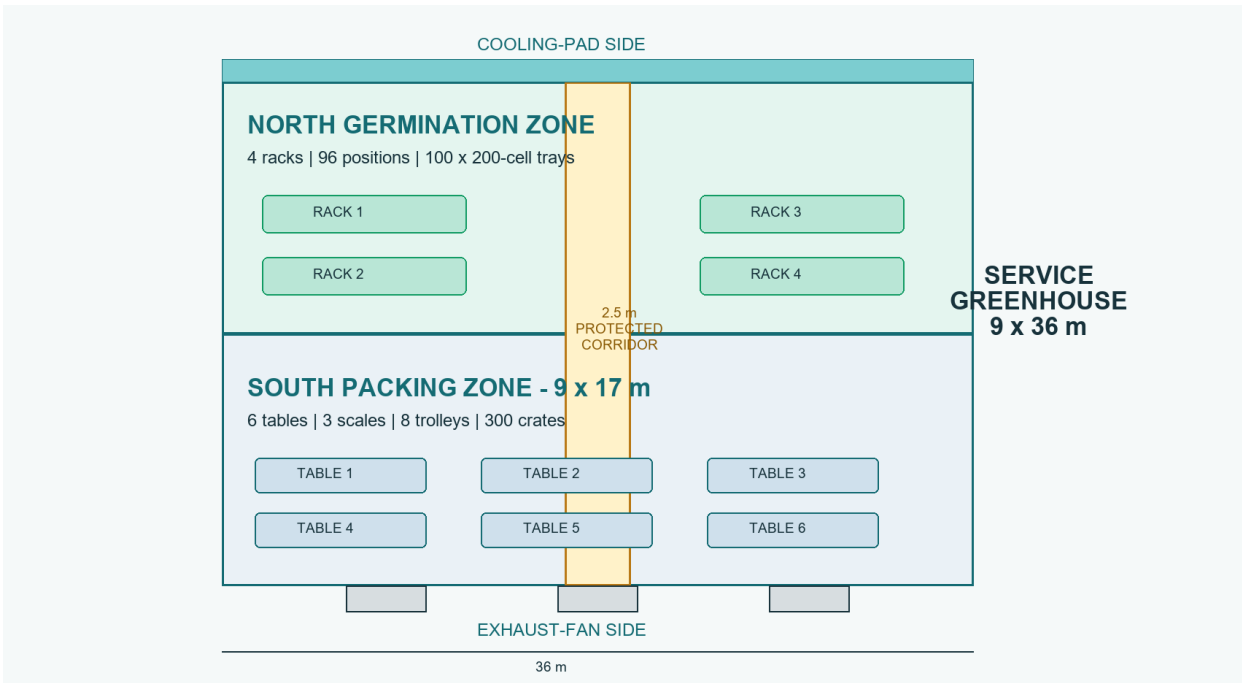


Figure 1 - Functional zoning. The protected 2.5 m corridor remains unobstructed; wet germination flow is separated from southern dry-packing flow.

### 3. Applicable Requirements and Submittals

- IEC 60598-1:2024 for luminaire safety and IEC 62471-7:2023 for photobiological safety.
- IEC 60529 and the project IEC 60364 electrical requirements for wet-area equipment.
- Codex CXC 1-1969 General Principles of Food Hygiene.
- Cornell CEA Lettuce Handbook for seedling PPFD and DLI verification.
- Approved supplier manuals, Egyptian statutory requirements and AFC layout; the stricter requirement governs.

Submit rack/load data, equipment data sheets, pump curve, nozzle data, piping schematic, PPFD data, electrical schedules, food-contact declarations, cleaning plan and commissioning procedures.

### 4. Germination Capacity and Equipment

| Code         | Equipment                     | Qty | Minimum requirement                                                                                       |
|--------------|-------------------------------|-----|-----------------------------------------------------------------------------------------------------------|
| GER-RCK-001  | Multi-level rack              | 4   | 2.4 x 0.65 x 2.1 m; 3 levels; 8 trays/level; galvanized or stainless moisture-resistant construction.     |
| GER-TRY-001  | 200-cell propagation tray     | 100 | Reusable agricultural plastic; 90 trays in normal operation; 96 physical rack positions.                  |
| GER-DRK-001  | Dark germination cabinet/rack | 1   | Minimum 100-tray capacity for first 24-48 h; moisture-resistant, cleanable and lightly ventilated.        |
| GER-TNK-001  | Independent germination tank  | 1   | 1 m3 food-grade PE tank inside the service greenhouse; latest approved value superseding earlier studies. |
| GER-PMP-001  | Independent pump              | 1   | 0.5 HP / 0.37 kW duty pump dedicated to the germination mist/spray circuit.                               |
| GER-TBL-001  | Tray preparation table        | 1   | SS304 or hygienic washable top, approximately 2.0 x 0.8 x 0.9 m high.                                     |
| GER-SNK-001  | Small washing sink            | 1   | SS304, approximately 600 x 500 mm, with tap, trapped drain and cleanable waste connection.                |
| GER-CAB-001  | Closed tool cabinet           | 1   | Moisture-resistant, approximately 1.0 x 0.45 x 1.8 m.                                                     |
| GER-TRL-001  | Tray trolley                  | 1   | Washable galvanized/stainless frame; waterproof castors; two brakes; 20-30 trays/trip.                    |
| GER-SCL-001  | Precision seed scale          | 1   | 3-5 kg capacity; preferred 0.01 g readability; not used for finished product.                             |
| GER-LOG-001  | Temperature/RH logger         | 2   | One at the dark cabinet and one between illuminated racks; exportable data.                               |
| GER-THM-001  | Tank thermometer              | 1   | Digital probe, minimum 0-50 degC range.                                                                   |
| GER-ECPH-001 | Handheld EC/pH meter          | 1   | For treated water and any agronomist-authorized light nutrient solution; calibration solutions included.  |

Four racks provide 96 tray positions. Normal loading is 90 trays and 18,000 seedlings per batch. Procurement includes 100 reusable 200-cell trays and 20,000 Rockwool plugs per cycle.

## 5. Local Water and Mist System

| Section        | Size/quantity                  | Complete installation requirement                                                                                 |
|----------------|--------------------------------|-------------------------------------------------------------------------------------------------------------------|
| WTP interface  | 1 in uPVC treated-water branch | 45 m route allowance from WTP outlet to the 1 m <sup>3</sup> germination tank; isolation at both ends.            |
| Pump suction   | Short flooded suction          | Tank outlet, full-bore isolation valve, union, strainer where required and eccentric/reducing connection to pump. |
| Pump discharge | 1/2 in local main              | Non-return valve, union, isolation valve, filter, pressure regulator, gauge and 24 V solenoid in flow direction.  |
| Local main     | 20 m of 1/2 in line            | Routes from pump assembly to four racks; clipped, drainable and accessible.                                       |
| Shelf laterals | 35 m of 16 mm line             | Twelve independent laterals, one per rack level, each with a small control valve and openable flush end.          |
| Spray heads    | 48 duty + 7 spare = 55         | Four removable cleanable light-spray nozzles per shelf level; uniform wetting without flooding.                   |
| Control        | Cycle timer + manual override  | Misting timer commands the 24 V solenoid/pump through interposing relay; low tank level prevents dry running.     |

### 5.1 Required flow order

Treated-water branch -> inlet isolation/air gap -> 1 m<sup>3</sup> tank -> outlet valve -> suction union -> 0.37 kW pump -> discharge union -> non-return valve -> isolation -> filter -> regulator -> gauge -> 24 V solenoid -> 1/2 in main -> four rack branches -> twelve 16 mm laterals -> valves -> nozzles -> flushable ends.

- Support lines independently from light cables and food-contact shelves.
- Flush all lines before installing final nozzle elements.
- Use treated water initially; any nutrient solution requires agronomist approval and recorded EC/pH.
- Prevent ponding, media washout and electrical overspray.

## 6. Grow-Lighting System

| Code                 | Item                                                                                                       | Qty                                                   | Technical requirement                                                                           |
|----------------------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| LGT-GRW-001          | Full-spectrum dimmable LED grow bar                                                                        | 24                                                    | 90 W each; IP65 minimum; two bars per shelf level; total connected load 2.16 kW.                |
| LGT-GRW-002          | Galvanized sheet-metal support/bracket set                                                                 | 24 sets                                               | Corrosion-resistant, height-adjustable, no sharp edge over trays, with secondary safety tether. |
| LGT-GRW-003          | Rack lighting control zones                                                                                | 4                                                     | One independently protected/timed zone per rack; local isolation and retained timer program.    |
| LGT-GRW-004          | Quantum/PAR acceptance measurement                                                                         | 1 test set                                            | Map PPFD at tray plane after rack loading and final mounting-height adjustment.                 |
| Growth stage         | Initial criterion                                                                                          | Commissioning action                                  |                                                                                                 |
| First 24 h           | Minimum 50 $\mu\text{mol}/\text{m}^2/\text{s}$ PPFD                                                        | Map and adjust mounting height/dimming.               |                                                                                                 |
| Seedling development | Adjustable up to 250 $\mu\text{mol}/\text{m}^2/\text{s}$ ; DLI about 22 $\text{mol}/\text{m}^2/\text{day}$ | Set photoperiod/dimming with agronomist and daylight. |                                                                                                 |
| Dark cabinet         | No grow lighting                                                                                           | Prevent light leakage; monitor T/RH.                  |                                                                                                 |

## 7. Germination Operating Method

| Step | Activity                          | Controlled execution                                                                                                    |
|------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| 1    | Seed and material receipt         | Check lot, cultivar, germination certificate, tray and Rockwool cleanliness; record traceability code.                  |
| 2    | Preparation and sowing            | Sanitize the table/tools, hydrate Rockwool uniformly, place one seed per cell and label each tray.                      |
| 3    | Dark germination                  | Hold trays in the clean dark cabinet for 24-48 h, avoiding excessive condensation and recording T/RH.                   |
| 4    | Rack transfer                     | Move emerged trays to the four illuminated racks; confirm zone, timer and nozzle alignment.                             |
| 5    | Misting and lighting              | Use short verified cycles; treated water initially; only agronomist-authorized light nutrient solution after emergence. |
| 6    | Inspection and grading            | Record germination percentage, uniformity, root condition, EC/pH and corrective action.                                 |
| 7    | Release to NFT nursery/production | Release only healthy, uniform, traceable seedlings; disinfect reusable trays before return.                             |

Batch records identify seed lot, sowing time, dark-cabinet period, rack/shelf, timer/light settings, T/RH, water temperature, EC/pH, germination percentage and release destination.

## 8. Southern Packing Area - 9 x 17 m

| Code                                                                                                                                                 | Equipment                            | Qty                                                                                                                 | Minimum requirement                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| SV-007                                                                                                                                               | Food-safe packing tables             | 6                                                                                                                   | SS304 preferred, or heavy-duty food-grade industrial plastic; smooth, washable, corrosion-resistant surfaces.                |
| SV-008                                                                                                                                               | Finished-product digital scales      | 3                                                                                                                   | Receiving, packing and final verification stations; cleanable platform; legal-for-trade approval if sold by declared weight. |
| SV-009                                                                                                                                               | Washable handling trolleys           | 8                                                                                                                   | Non-marking wheels, stable load platform and brakes where stationed.                                                         |
| SV-010                                                                                                                                               | Food-grade harvesting/packing crates | 300                                                                                                                 | Ventilated, stackable, washable and dedicated to clean produce handling.                                                     |
| Step                                                                                                                                                 | Operation                            | Hygiene/control requirement                                                                                         |                                                                                                                              |
| 1                                                                                                                                                    | Receive                              | Move harvested lettuce in clean food-grade crates through the protected service route.                              |                                                                                                                              |
| 2                                                                                                                                                    | Inspect and grade                    | Remove damaged or contaminated product; separate grades without floor contact.                                      |                                                                                                                              |
| 3                                                                                                                                                    | Trim/dry clean                       | Current basis is dry handling only; use clean dedicated tools and prevent product contact with waste.               |                                                                                                                              |
| 4                                                                                                                                                    | Weigh                                | Use the three finished-product scales at receiving, packing and final verification points.                          |                                                                                                                              |
| 5                                                                                                                                                    | Pack and identify                    | Pack manually into approved food-contact packaging and apply lot/date identification required by the sales channel. |                                                                                                                              |
| 6                                                                                                                                                    | Dispatch                             | Move packed product to dispatch promptly; protect from sun, fan discharge, chemicals and dirty returns.             |                                                                                                                              |
| CURRENT PACKING BOUNDARY                                                                                                                             |                                      |                                                                                                                     |                                                                                                                              |
| Dry manual handling only. No produce washer, process-water line, automatic sealer, label printer or cold room is included in the current base scope. |                                      |                                                                                                                     |                                                                                                                              |

## 9. Installation Sequence

- Approve coordinated layout, airflow, corridor, doors, drains and service points.
- Accept concrete floor, falls, drains, sleeves and protected route.
- Install, level and stabilize four racks and all cleanable shelves.
- Install the 1 m3 tank, 0.37 kW pump and 1 in/1/2 in/16 mm water system in the defined order.
- Flush, wet-test and balance 48 operating nozzles; retain seven spares.
- Install 24 LED bars, galvanized brackets, tethers, four control zones and local isolation.
- Install preparation, hygiene and monitoring equipment.
- Install six tables, three scales, eight trolleys and clean crate storage for one-way flow.
- Complete cleaning, tests, PPFD mapping, trial batch and training.

## 10. Hygiene, HSE and Environmental Controls

| Hazard              | Mandatory control                                                                 | Risk |
|---------------------|-----------------------------------------------------------------------------------|------|
| Cross-contamination | Protected access, disinfection, cleanable surfaces, batch separation and records. | Low  |
| Water/electricity   | IP65, 24 V solenoid, RCD, sealed glands, drip loops and clearance.                | Low  |
| Rack overload       | Approved loads, level base, anchors where required and load labels.               | Low  |
| Slips               | Positive falls, trapped drain and prompt cleanup.                                 | Low  |
| Light exposure      | Risk classification and isolation before servicing.                               | Low  |
| Manual handling     | Trolleys, crate limits and clear aisles.                                          | Low  |
| Chemicals           | Locked labeled storage away from open trays and produce.                          | Low  |

## 11. Testing, Commissioning and Acceptance

| Test                | Method                                                    | Acceptance                                                                                |
|---------------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Layout and access   | Measure racks, tables, aisles and protected corridor      | No obstruction; safe trolley movement; doors and cooling airflow remain clear.            |
| Rack proof/load     | Apply not less than 1.25 x normal wet operational load    | No instability, permanent distortion or loose fastener.                                   |
| Tank and piping     | Fill, inspect and pressure/leak test                      | No leakage; tank stable; all valves accessible and correctly labeled.                     |
| Pump duty           | Operate through all 12 laterals                           | Stable pressure, no cavitation, dry-run trip proven and current within nameplate.         |
| Nozzle uniformity   | Collect discharge from representative/all nozzles         | No blocked heads; visually uniform wetting; each level balances by valve.                 |
| Lighting electrical | Insulation, earth continuity, RCD and functional test     | Pass electrical method limits; each rack isolates and times independently.                |
| PPFD map            | Grid measurement at tray plane                            | Minimum 50 umol/m2/s during first 24 h; adjustable recipe up to 250 umol/m2/s thereafter. |
| Drainage            | Wet floor/sink test                                       | Positive flow to trapped drain; no ponding or backflow; NFT return remains isolated.      |
| Packing hygiene     | Cleanability and mock product-flow inspection             | No dirty/clean cross-flow, floor contact, chemical exposure or uncleanable surface.       |
| Integrated trial    | One complete sowing-to-release batch and packing mock run | Traceable records, stable conditions and trained operators.                               |

### 11.1 Inspection and Test Plan

| No. | Inspection stage                        | Method                 | Type | Record                 |
|-----|-----------------------------------------|------------------------|------|------------------------|
| 01  | Coordinated service layout              | Drawing review         | H    | Approved layout        |
| 02  | Rack shop drawings and load data        | Document review        | H    | Fabrication release    |
| 03  | Tank/pump/mist materials                | Certificate and visual | W    | Material release       |
| 04  | Concealed supports, drains and services | Inspection             | H    | Close-up release       |
| 05  | Rack, table and equipment installation  | Survey/visual          | W    | Installation checklist |
| 06  | Piping flush, leak and pump test        | Functional             | H    | Wet-test record        |
| 07  | Lighting electrical and PPFD test       | Instrumented           | H    | Electrical/PPFD report |
| 08  | Hygiene and packing workflow            | Mock run               | W    | GHP checklist          |
| 09  | Integrated operating batch              | Performance            | H    | Commissioning log      |
| 10  | Training, O&M and records               | Record review          | R    | Handover dossier       |

H = Hold, W = Witness, R = Record review. HP-SRV-07 covers lighting electrical and PPFD acceptance; HP-SRV-09 covers the integrated operating batch. No production release is permitted before both hold points are accepted.

## 12. Handover and Training

- Approved layout, rack/load data, tank/pump/nozzle data and as-builts.
- Wet tests, pump/nozzle balance, electrical tests, PPFD maps, calibration and trial records.
- Cleaning, traceability, rejection and waste procedures.
- O&M manuals, warranties, nozzle spares, calibration materials and supplier contacts.
- Operator training for sowing, dark germination, rack transfer, settings, measurements, cleaning, weighing, packing, alarms and isolation.

### 12.1 Public references

IEC 60598-1:2024: <https://webstore.iec.ch/en/publication/66620> | IEC 62471-7:2023: <https://webstore.iec.ch/en/publication/68810> | Codex CXC 1-1969: <https://www.fao.org/fao-who-codexalimentarius/codex-texts/codes-of-practice/en/> | Cornell CEA Lettuce Handbook: <https://cea.cals.cornell.edu/files/2019/06/Cornell-CEA-Lettuce-Handbook-.pdf>